

STUDENT'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

DECEMBER ASSESSMENT

2007

MATHEMATICS

EXTENSION 2

GENERAL INSTRUCTIONS:

- Time allowed - 70 minutes.
- Write using blue or black pen.
- Start each question on a new page.
- Write your name at the top of each page.
- Calculators may be used
- **ALL** necessary working should be shown in every Question.

QUESTION 1 (16 marks) Start on a new page

Marks

- (a) If $z_1 = 5 - 2i$ and $z_2 = i - 4$ express the following in the form $a + ib$ where a and b are real numbers.

i) $z_1 + z_2$ 1

ii) $z_1 z_2$ 2

iii) $\frac{z_1}{z_2}$ 2

- b) Sketch the region in the complex plane where the inequalities:

$$|z - 1 - 2i| < 5 \quad \text{and} \quad 0 < \arg(z - 1 - 2i) < \frac{\pi}{3} \quad \text{both hold} \quad \text{3}$$

- (c) If $z = a + ib$ where a and b are real numbers, find

i) $\text{Im}(4iz - 3)$ 2

ii) $\overline{3iz}$ in the form $x + iy$ where x and y are real numbers 3

- (d) Find a and b such that

$$(1 + i\sqrt{3})^8 = a + ib \quad \text{3}$$

QUESTION 2 (17 marks) Start on a new page

Marks

- (a) Find $\sqrt{6i-8}$ and hence solve the quadratic equation: 4

$$2z^2 - (3+i)z + 2 = 0 \quad \text{expressing your answer in the form}$$

$$x + iy$$

- (b) If ω is a complex root of the equation:

$$z^5 - 1 = 0$$

- i) Show that ω^2, ω^3 and ω^4 are the other complex roots. 2

- ii) Prove that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$ 1

- iii) Form the quadratic equation with roots 3

$$(\omega^2 + \omega^3) \quad \text{and} \quad (\omega + \omega^4)$$

- (c) A represents the complex number $2 + 3i$ in the complex plane

Find a possible complex number represented by B such that

triangle OAB is a right angled isosceles triangle with right angle at

- i) O 1

- ii) A 1

- iii) B 2

- (d) Given $z = \frac{1}{2}(\cos\theta + i\sin\theta)$ where θ is real, show that the imaginary 3

part of $\frac{1}{1-z}$ is

$$\frac{2\sin\theta}{5-4\cos\theta}$$

QUESTION 3 (15 marks) Start on a new page

Marks

- (a) By letting $z = \cos\theta + i\sin\theta$ and using De Moivre's Theorem, find an expression for both $\sin 3\theta$ and $\cos 3\theta$ and hence show that

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

- (b) i) If $z = \cos\theta + i\sin\theta$ show that: 3

$$z + z^{-1} = 2\cos\theta \quad \text{and}$$

$$z^n + z^{-n} = 2\cos n\theta \quad \text{and find corresponding expressions for}$$

$$z - z^{-1} \quad \text{and} \quad z^n - z^{-n}$$

- ii) By expanding $\left(z + \frac{1}{z}\right)^4$ find an expression for $\cos^4\theta$ in the form 2

$$A\cos 4\theta + B\cos 2\theta + C$$

- c) Find the equation of the locus of z if 3

$$\text{Arg}\left(\frac{z+1}{z-3}\right) = \frac{\pi}{2} \quad \text{stating any restrictions.}$$

- d) By considering the triangle inequality 3

$$|z_1 - z_2| \geq |z_1| - |z_2| \quad \text{find the greatest value of } |z| \text{ which is}$$

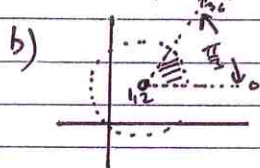
satisfied by the equation:

$$\left|z - \frac{7}{z}\right| = 4$$

END OF EXAMINATION

1 a) i) $1-i$
 ii) $(5-2i)(i-4)$
 $= 5i - 20 - 2i^2 + 8i$
 $= 13i - 18$
 iii) $\frac{5-2i}{-(4+i)} \times \frac{4-i}{4-i}$
 $= -\frac{(20-5i-8i+2i^2)}{17}$

$$= \frac{-18 + 13i}{17}$$



c) i) $4iz - 3$
 $= 4i(a+ib) - 3$
 $= 4ia - 4b - 3$
 $\therefore \operatorname{Im}(4iz - 3) = 4a$
 ii) $3iz = 3i(a+ib)$
 $= 3ai - 3b$
 $\overline{3iz} = -3b - 3ai$

d) $(1+i\sqrt{3}) = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$
 $= 2 \operatorname{cis} \frac{\pi}{3}$
 $(1+i\sqrt{3})^8 = 2^8 \operatorname{cis} 8\frac{\pi}{3}$
 $= 2^8 \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$
 $= -128 + 128i\sqrt{3}$

2 a) Let $z = x+iy = \sqrt{6i-8}$

$$x^2 - y^2 + 2xyi = 6i - 8$$

$$x^2 - y^2 = -8$$

$$2xy = 3$$

$$\text{Let } y = \frac{3}{x}$$

$$\therefore x^2 - \frac{9}{x^2} = -8$$

$$x^4 + 8x^2 - 9 = 0$$

$$(x^2+9)(x^2-1) = 0$$

$$\therefore x = \pm 1 \quad y = \mp 3$$

$$\therefore \sqrt{6i-8} = \mp(1+3i)$$

$$23^2 = -(3+i)z + 2 = 0$$

$$z = \frac{3+i \mp \sqrt{9+6i-1-16}}{4}$$

$$= \frac{3+i \mp \sqrt{6i-8}}{4}$$

$$= \frac{3+i \mp (1+3i)}{4}$$

$$= \frac{4+4i}{4} \text{ or } \frac{2-2i}{4}$$

$$= (1+i) \text{ or } \frac{1}{2}(1-i)$$

ii) $z^5 - 1 = 0$

$$(z-1)(z^4+z^3+z^2+z+1) = 0$$

$$(z-1)(1+z+z^2+z^3+z^4) = 0$$

if w is a root $w \neq 1$ as complex

$$\therefore 1+w+w^2+w^3+w^4 = 0$$

Can also do it by Σ

$$\Sigma d = 1 \quad \therefore 1+w+w^2+w^3+w^4 = 0$$

iii) $\alpha + \beta = w^2 + w^3 + w + w^4 = -1$

$$\alpha\beta = (w^2+w^3)(w+w^4)$$

$$= w^3 + w^6 + w^4 + w^7$$

$$= w^3 + w + w^4 + w^2 = -1$$

$$\therefore \text{quad in } z^2 + z - 1 = 0.$$

c) i) $i(2+3i)$ or $-i(2+3i)$
 $= -3+2i$ $3-2i$

ii) $(-3+2i) + (2+3i)$ or $(3-2i) + (2+3i)$
 $-1+5i$ $5+i$

iii) $\frac{1}{2}(-1+5i)$ or $\frac{1}{2}(5+i)$
 $-\frac{1}{2} + \frac{5i}{2}$ $\frac{5}{2} + \frac{i}{2}$

b) $z^5 - 1 = 0$

if w is a root

$$w^5 - 1 = 0$$

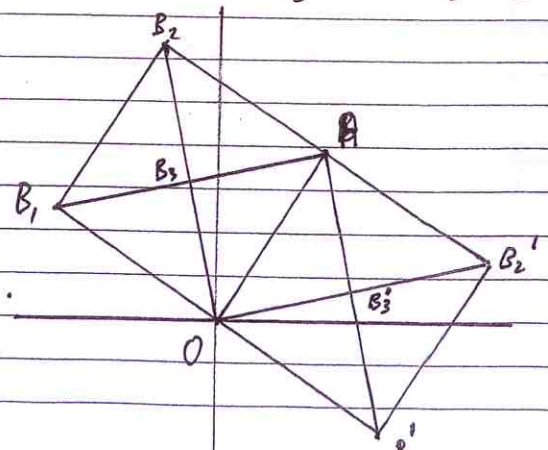
$$w^5 = 1$$

$$\text{then } (w^2)^5 = (w^5)^2 = 1$$

$$(w^3)^5 = (w^5)^3 = 1$$

$$(w^4)^5 = (w^5)^4 = 1$$

$\therefore w^2, w^3, w^4$ are also roots.



$$2) d) \quad z = \frac{1}{2}(\cos \theta + i \sin \theta)$$

$$\frac{1}{1-z} = \frac{1}{1 - \frac{1}{2}(\cos \theta + i \sin \theta)}$$

$$= \frac{2}{2 - \cos \theta - i \sin \theta}$$

$$= \frac{2}{(2 - \cos \theta) - i \sin \theta} \times \frac{(2 - \cos \theta) + i \sin \theta}{(2 - \cos \theta) + i \sin \theta}$$

$$= \frac{(4 - 2\cos \theta) + 2i \sin \theta}{4 - 4\cos \theta + \cos^2 \theta + \sin^2 \theta}$$

$$= \frac{(4 - 2\cos \theta) + 2i \sin \theta}{5 - 4\cos \theta}$$

$$\therefore \operatorname{Im}\left(\frac{1}{1-z}\right) = \frac{2 \sin \theta}{5 - 4 \cos \theta}$$

$$3) a) \quad z = \cos \theta + i \sin \theta$$

$$(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + i \sin^3 \theta$$

$$\therefore \cos^3 \theta + i \sin^3 \theta = \cos^3 \theta + 3\cos^2 \theta i \sin \theta + 3\cos \theta i^2 \sin^2 \theta + i^3 \sin^3 \theta$$

$$= (\cos^3 \theta - 3\cos \theta \sin^2 \theta) + (3\cos^2 \theta \sin \theta - \sin^3 \theta)i$$

$$\therefore \sin^3 \theta = 3\cos^2 \theta \sin \theta - \sin^3 \theta \quad \cos^3 \theta = \cos^3 \theta - 3\cos \theta \sin^2 \theta$$

$$= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta$$

$$= \cos^3 \theta - 3\cos \theta (1 - \cos^2 \theta)$$

$$= 3 \sin \theta - 3 \sin^3 \theta - \sin^3 \theta$$

$$= \cos^3 \theta - 3\cos \theta + 3\cos^3 \theta$$

$$= 8 \sin \theta - 4 \sin^3 \theta$$

$$= 4\cos^3 \theta - 4\cos \theta$$

$$4 \sin^3 \theta = 8 \sin \theta - 4 \sin^3 \theta$$

$$= \frac{3\cos^2 \theta \sin \theta - \sin^3 \theta}{\cos^3 \theta - 3\cos \theta \sin^2 \theta}$$

$\div$ both by $\cos^3 \theta$

$$\therefore \tan^3 \theta = \frac{3\cos^2 \theta \sin \theta}{\cos^3 \theta} - \frac{\sin^3 \theta}{\cos^3 \theta}$$

$$= \frac{\cos^2 \theta}{\cos^3 \theta} - \frac{3\cos \theta \sin^2 \theta}{\cos^3 \theta}$$

$$= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

$$b) \quad i) \quad z = \cos \theta + i \sin \theta$$

$$z^{-1} = \cos(-\theta) + i \sin(-\theta)$$

$$= \cos \theta - i \sin \theta$$

$$\therefore z + z^{-1} = 2 \cos \theta$$

$$z - z^{-1} = 2i \sin \theta$$

$$z^n = \cos n\theta + i \sin n\theta$$

$$z^{-n} = \cos n\theta - i \sin n\theta$$

$$\therefore z^n + z^{-n} = 2 \cos n\theta$$

$$z^n - z^{-n} = 2i \sin n\theta$$

$$ii) \quad \left(z + \frac{1}{z}\right)^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$$

$$= \left(z^4 + \frac{1}{z^4}\right) + 4\left(z^2 + \frac{1}{z^2}\right) + 6$$

3) and

$$(2\cos\alpha)^4 = 2\cos 4\alpha + 4(2\cos 2\alpha) + 6$$

$$16\cos^4\alpha = 2\cos 4\alpha + 8\cos 2\alpha + 6$$

$$\cos^4\alpha = \frac{1}{8}\cos 4\alpha + \frac{1}{2}\cos 2\alpha + \frac{3}{8} \quad \perp$$

c) $\arg(3+i) - \arg(3-3i) = \frac{\pi}{2}$

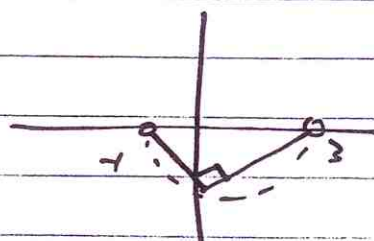
min circle $y < 0$ with

diameter between $(-1, 0)$ $(3, 0)$ $\perp$

center is $(1, 0)$ radius = 2

$$\therefore (x-1)^2 + (y-0)^2 = 4 \quad \perp$$

$$(x-1)^2 + y^2 = 4 \quad y < 0$$



d) $\left| 3 - \frac{7}{3} \right| \geq \left| 3 \right| - \frac{7}{|3|}$

$$\therefore \left| 3 \right| - \frac{7}{|3|} \leq 4 \quad \perp$$

$$\left| 3 \right|^2 - 7 \leq 4\left| 3 \right|$$

$$\left| 3 \right|^2 - 4\left| 3 \right| - 7 \leq 0$$

if $\left| 3 \right|^2 - 4\left| 3 \right| - 7 = 0$

$$\left| 3 \right| = \frac{4 \pm \sqrt{16 + 28}}{2}$$

$$= 2 \pm \sqrt{11}$$

$$\therefore 2 - \sqrt{11} \leq \left| 3 \right| \leq 2 + \sqrt{11}$$

$$\therefore \min \left| 3 \right| = 2 - \sqrt{11} \quad \perp$$

